

虚短: $V_1 = 0V$

$$i_1 = \frac{2V}{1k} = 2mA$$

$$i_2 = i_1 = 2mA$$

$$V_0 = V_1 - i_2 \times 5k = -10V$$

$$i_L = \frac{-10}{1k} = -10mA$$

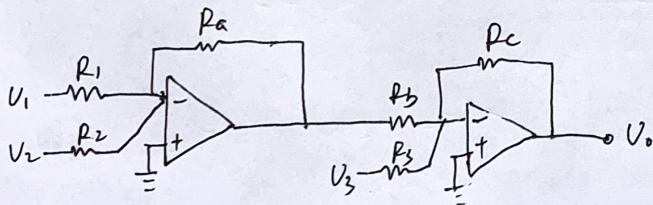
$$\rightarrow i_0 = i_L - i_2 = -10 - 2 = -12mA$$

$$\frac{V_0}{V_i} = \frac{-10}{2} = -5 = -13.98dB$$

$$\frac{i_L}{i_1} = -5 = -13.98dB$$

$$\frac{P_0}{P_i} = \frac{V_0}{V_i} \cdot \frac{i_L}{i_1} = 25 = 13.98dB$$

② 2.8 Inverting config 增益为负. $\therefore R_2, R_4$ 实现“负”; R_1, R_3 实现“正”



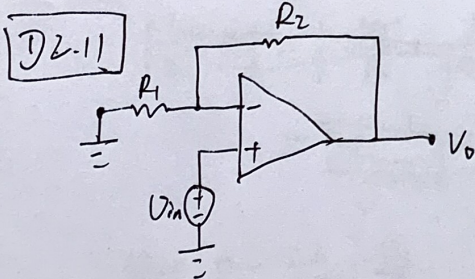
$$V_0 = 2V_1 + V_2 - 4V_3$$

10kΩ 是常规电阻. 将 R_a, R_b, R_c 设为 10kΩ. 调节 R_1, R_2, R_3 实现加减运算.

$$-4V_3 \Rightarrow \frac{R_c}{R_3} = 4 \Rightarrow R_3 = 2.5k\Omega$$

$$2V_1 \Rightarrow \frac{R_a}{R_1} = 2 \Rightarrow R_1 = 5k\Omega$$

$$V_2 \Rightarrow \frac{R_b}{R_2} = 1 \Rightarrow R_2 = 10k\Omega$$



$$1 + \frac{R_2}{R_1} = 2 \Rightarrow R_1 = R_2$$

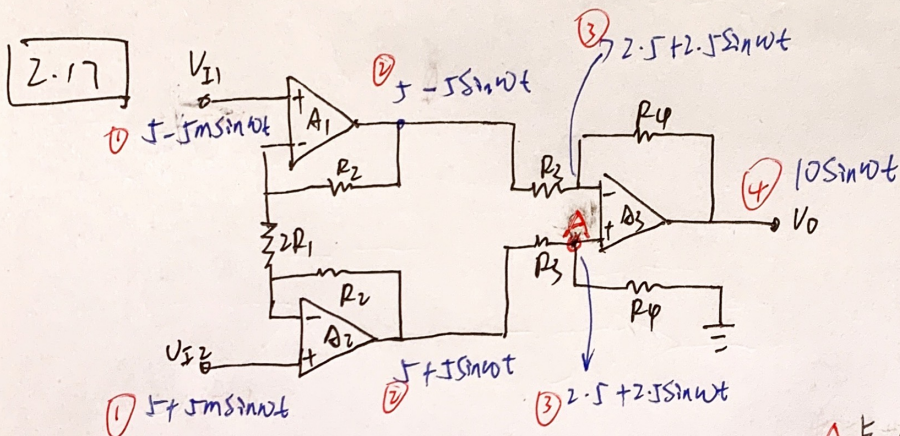
$$\frac{10V}{R_1 + R_2} = 10\mu A \Rightarrow R_1 + R_2 = 1m$$

$$\therefore R_1 = R_2 = 0.5mA$$



D2.16 $R_1 = R_3$; $R_2 = R_4$

$$\begin{cases} \frac{R_2}{R_1} = 10 \\ 2R_1 = 20k \end{cases} \Rightarrow \begin{cases} R_1 = 10k = R_3 \\ R_2 = 100k = R_4 \end{cases}$$



stage 1 的增益

$$1 + \frac{R_2}{R_1} = 1 + \frac{0.5M}{0.5k} = 1000$$

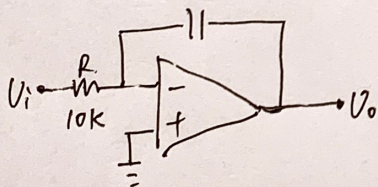
1001 可近似为 1000

stage 2 的增益

$$\frac{R_4}{R_3} = 1$$

A 点处为 R_3 、 R_4 的分压

D2.19



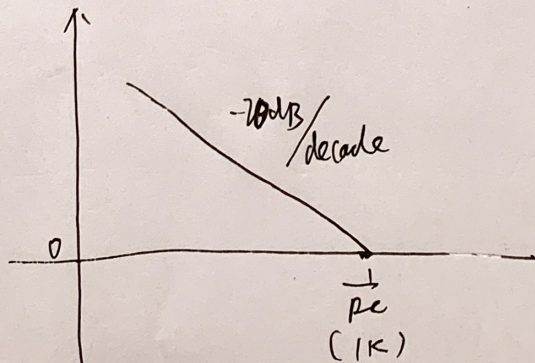
$$R_c = 10^{-3} s, R = 10k \Rightarrow C = 0.1 \mu F$$

$$\omega_c = \frac{1}{R_c} = 1k \text{ rad/s}$$

$$\begin{aligned} \left| \frac{V_o}{V_i}(\omega) \right| &= \left| \frac{1}{R} \cdot \frac{1}{j\omega C} \right| \\ &= \left| \frac{1}{j\omega R C} \right| \end{aligned}$$

$$\begin{aligned} V_o &= 0 - \frac{1}{j\omega C} \cdot \frac{V_i}{R} \\ &= j \frac{1}{\omega R C} \cdot V_i \end{aligned}$$

$$\frac{V_o}{V_i} = j \frac{1}{\omega R C}$$



$$\therefore 100 \text{ rad/s} \rightarrow 20 \text{ dB}$$

$$10 \text{ rad/s} \rightarrow 40 \text{ dB} \Rightarrow 100 \text{ V/V}$$

$$1 \rightarrow 60 \text{ dB} \Rightarrow 1000 \text{ V/V}$$

相位为 90°